Muhammad Uzair

AI Engineer

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I build AI that people talk to. Five years in AI and machine learning, most recently building Aida, an AI financial adviser, where the answers had to be right as well as fast. I like problems where the model is only half the job; the rest is making it quick, keeping the data honest, and getting it into people's hands.

SKILLS

AI & LLMs: Agent architecture, tool use, RAG, embeddings and hybrid search, context and memory management, evals (regression suites and LLM-as-judge), prompt caching

Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, NLP

Frameworks & Platforms: LangGraph, LangChain, CrewAI, Microsoft Agent Framework, Azure AI Foundry, Azure AI Search, OpenAI models, Claude Code

Engineering: Python, C#, Docker, AWS (ECS, CDK, SageMaker), Redis, PostgreSQL, MongoDB, OpenTelemetry

PROFESSIONAL EXPERIENCE

AI Engineer

Money Means *08/2025 – 10/2026*

Context: Aida was an AI financial adviser. Responsible for the parts of it that understood what a user said, remembered it, and used it in later conversations. Money Means closed in October 2026.

Specific Tasks

- Owned the language-model agent behind Aida's chat: tool selection, conversation flow, and how facts a user shared were captured and reused.
- Replaced a multi-call extraction loop with a single-call design: **3× faster on average, 5× faster at the tail**, no loss in accuracy.
- Designed the step that decided which tools the model saw for each message. Correct tool choice rose from **88% to 97%**.
- Introduced prompt caching across all model calls by separating the fixed and per-turn parts of each prompt.
- Set the patterns the team built agent tools against: a typed contract per tool, structured JSON output per model call, and one place that defined what each tool needed before it could run.
- Designed how the agent read and wrote financial data: profile writes were typed and sat behind a confidence threshold; tools were gated on what the user's accounts contained; the model never wrote unvalidated data.
- Owned what the model saw each turn (history, rolling summary, user profile) so accuracy held as conversations grew.
- Routed models per task: a small model for classification and tool selection, a larger one for the reply, with automatic failover. Migrated services between model generations without regressions.
- Built automated regression testing for the agent: fixed conversation datasets with expected tool calls and outputs, run on every prompt, tool and model change.
- Built a prompt-assembly visualiser that showed exactly what the model saw on any turn.

Technologies: C#, Python, Pydantic, CrewAI, Microsoft Agent Framework (Microsoft.Agents.AI), Azure AI Search, Azure AI Foundry, Redis, OpenTelemetry

Full Stack Data Scientist

VUE group *11/2023 – 07/2025*

Context: Data science and backend work for a vehicle fleet business: research, proof-of-concept builds and analytics for internal teams.

Specific Tasks

- Developed and deployed conversational data visualisation services using LLMs and Dash, enabling non-technical users to access real-time analytics through natural language queries.
- Evaluated new AI tools and wrote research briefs that senior leadership used to decide what to invest in.
- Built a vehicle utilisation and monitoring system that cut manual reporting by 70%.
- Ran statistical modelling and exploratory analysis across MSSQL, PostgreSQL and MongoDB.
- Migrated databases from MSSQL to PostgreSQL with no data loss, improving query performance.
- Managed deployment pipelines with CircleCI and Docker, reducing release times.

Technologies: Python, AWS, LLMs, NLP, TensorFlow, Docker, Bitbucket, n8n, Postman, Hugging Face, LangChain, CircleCI, MSSQL, PostgreSQL, MongoDB

Co-founder

Innervoices S.L. *02/2023 – 11/2023*

Context: Co-founded a start-up building a web product that helped people talk through personal issues from both an emotional and a rational perspective. Owned the backend and deployment side in a small team. Paused in late 2023 when funding ran out.

Specific Tasks

- Built Flask REST APIs backed by MongoDB, containerised them with Docker and deployed them on AWS Fargate with AWS CDK, with SSL for HTTPS.
- Used open-source language models for the product's AI features, and curated the datasets behind them.
- Built **Retrieval-Augmented Generation** with Qdrant and Pinecone, so the product's answers were grounded in retrieved sources.

Technologies: Python, React, MongoDB, VectorDB, AWS, Docker, GitLab, Postman, pytest, Hugging Face, TensorFlow, LangChain, PyCharm, Google Analytics, Scikit-learn, SageMaker

Data Scientist

Plenty, Barcelona *10/2021 – 04/2022*

Used Python, machine learning and deep learning to speed up data collection and research at an early-stage start-up.

Specific Tasks

- Developed a Python web scraper that collected over half a million descriptions and stock quotes.
- Designed and built the start-up's machine-learning models on my own, and tested CNN image models (VGG-16, Inception).

Technologies: Python, SageMaker, Google Colab, GitHub, TensorFlow, Hugging Face

System Support Engineer

Techaccess Pakistan *01/2018 – 10/2020*

Provided technical support and IT services to clients, specialising in Oracle databases and Linux systems.

Specific Tasks

- Set up, administered and tuned Oracle RDBMS and Linux systems for client environments.
- Ran database backup and recovery using RMAN and Data Pump.

Technologies: Oracle RDBMS, Red Hat Linux, SQL, VMware, Fusion Middleware

EDUCATION

MSc Artificial Intelligence & Applications

University of Strathclyde *09/2020 – 09/2021*

- **Thesis:** NLP sentiment analysis using deep learning and transfer learning to work around limited training data.

BSc Electrical Engineering

National University of Computer and Emerging Sciences *09/2012 – 09/2016*

LANGUAGES

English (fluent), Urdu and Pashto (native)